Metrics

Numbers to quote on defense. Every figure is from a run recorded under docs/, not an estimate, except the one row marked as an estimate. The source of each table is named so the number can be reproduced.

Corpus

Counts from reading each capture with the capture engine (read_capture / build_sessions). The corpus is produced by scripts/generate_corpus.py and is reproducible from the committed scripts.

Capture	File	Packets	Sessions	Role
corpus_capture_01.pcapng	28.4 KiB	295	3	primary; 280 framed messages, 140 requests
corpus_capture_02.pcapng	8.3 KiB	90	2	held out from framing; used for applicability
corpus_capture_defects.pcapng	1.7 KiB	18	1	retransmission, out-of-order, gap, conflicting overlap
synthetic_live.pcapng	2.5 KiB	27	1	real TCP stack, different layout; applicability boundary

Real corpus

Source: docs/REAL_CORPUS_ANALYSIS.md, from the organizer dataset in tests/corpus/real/. Four research captures, five TCP sessions, 1793 stream bytes, 90 framed messages (45 requests, 45 responses).

Metric	Value
Sessions	5 (matches dataset_manifest.json)
Request messages	45, all matched, 100 % byte coverage
Response messages	45, all matched, 100 % byte coverage
Precision (matched / framed)	1.00
Counterexamples, rule v1	6 on the configuration capture (2 mismatched, 4 incomplete); 15 over the four research captures (5 mismatched, 10 incomplete)
Counterexamples, rule v2	0
Journal correlation	14 of 14 requests correlated on the configuration capture
Diagnostic captures	8, each reports its condition (gap, ambiguity, checksum_offload)

Command surface

Stage	Command	Values
Capture	`python -m src.capture.cli --pcap <f> --out <json>`	read
Project	`python -m src.project.cli pipeline --pcap <f> --project <d> --report <f>`	read, measure
Project	`python -m src.project.cli create | open | add-capture | export | import`	read, write
Protocol	`python -m src.protocol.cli apply --rule <f> --capture <f> --out <f>`	read, write
Report	`pipeline --report <f>` (Markdown and HTML; HTML off with `--no-html`; no standalone CLI)	write

Processing

Source: `docs/BENCHMARK.md`, produced by `scripts/run_benchmark.py` over a 250000-packet, 1000-session, 95.66 MiB capture on 4 vCPU / 16 GiB RAM.

Phase	Seconds
Parse	7.29
Sessions	0.70
Reassembly	1.84
Export	2.27
Total	12.10

Input	Packets	Sessions	Regulars	Regular peak MiB	Streamings	Streaming peak MiB
9.57 MiB	25000	100	1.10	58	1.06	54
47.84 MiB	125000	500	6.10	186	5.71	167
90.88 MiB	237500	950	11.46	336	10.86	293

Peak memory at the reference profile is 350 MiB against a 4 GiB budget; total time is 12.10 s against a 5 minute budget.

Rules

Source: `docs/REFERENCE_INVESTIGATION.md` (protocol engine). Rule v1 and rule v2 describe the same bytes; only the description changed.

Metric	v1	v2	Delta
matched	60	140	+80
mismatched	80	0	-80
counterexamples	80	0	-80
coverage	0.500000	0.500000	0.0
precision	0.428571	1.000000	+0.571429
counterexample density	57.142857	0.0	-57.142857

Framing candidates, all 10 streams: `payload` frames every stream with no leftover bytes; `payload_and_length_field` and `entire_message` do not.

Candidate	Streams framed cleanly
payload	10 / 10
payload and length field	0 / 10
entire message	0 / 10

Counterexamples

Rule	Counterexamples in the corpus
v1	80 (60 command bytes of value `2`, 20 of value `3`, none in the expected set `[1]`)
v2	0

Resolved by the v1 -> v2 change: 80. Introduced: 0.

Journal correlation

Source: `docs/REFERENCE_INVESTIGATION.md`.

Capture	Requests	Correlated	Window
corpus_capture_01.pcapng	140	140	500 ms
corpus_capture_02.pcapng	40	40	500 ms

Robustness

Damaged and unusual captures are processed without an uncaught error; each becomes a diagnostic or a clearly marked unknown.

Fixture	Condition
defects/fragmented_ip.pcap	IP fragment
defects/mixed_ip_versions.pcapng	IPv4 and IPv6 in one file
defects/mixed_link_types.pcapng	two link types in one file
defects/truncated_header.pcapng	header cut short
defects/truncated_packet.pcapng	payload cut short
defects/unsupported_link_type.pcapng	link type the parser does not know
ipv6_mixed.pcapng	IPv6 present
no_syn.pcapng	no SYN, roles unknown
overlap_conflict.pcapng	conflicting overlap

The defect capture (18 packets) normalizes in 0.021 s regular and 0.022 s streaming, so diagnostics do not degrade throughput at this size.

Manual comparison

Approach	Time for 100 messages
Manual reading in a hex editor	4-6 hours (estimate)
Rule applied to the corpus	seconds

The manual figure is an estimate for the defense narrative and is not a measurement; the automated figure is a run.